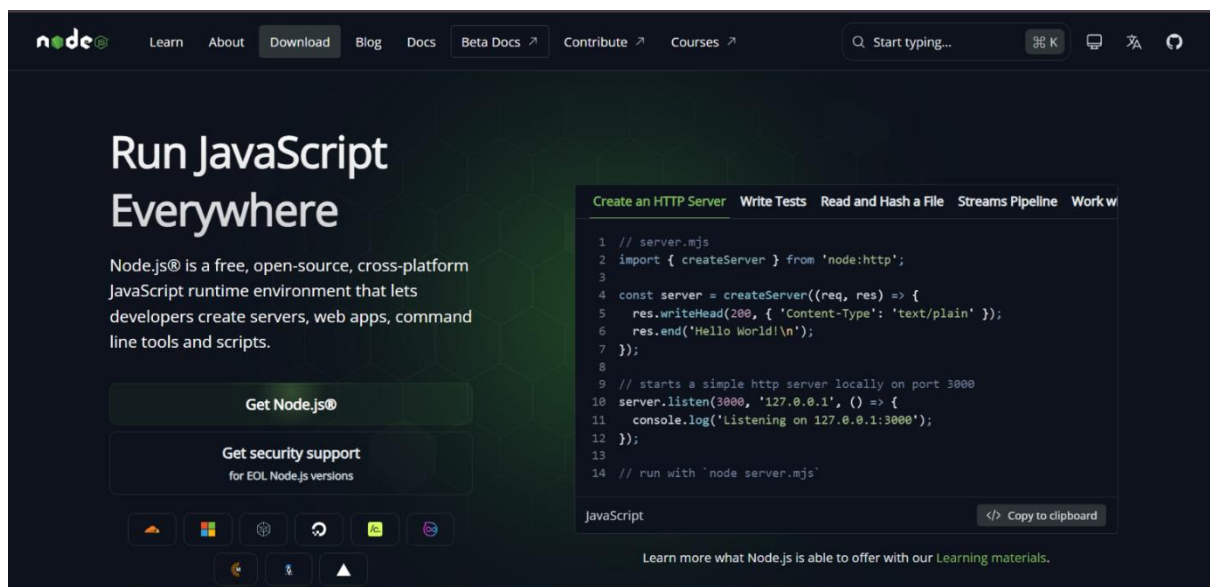
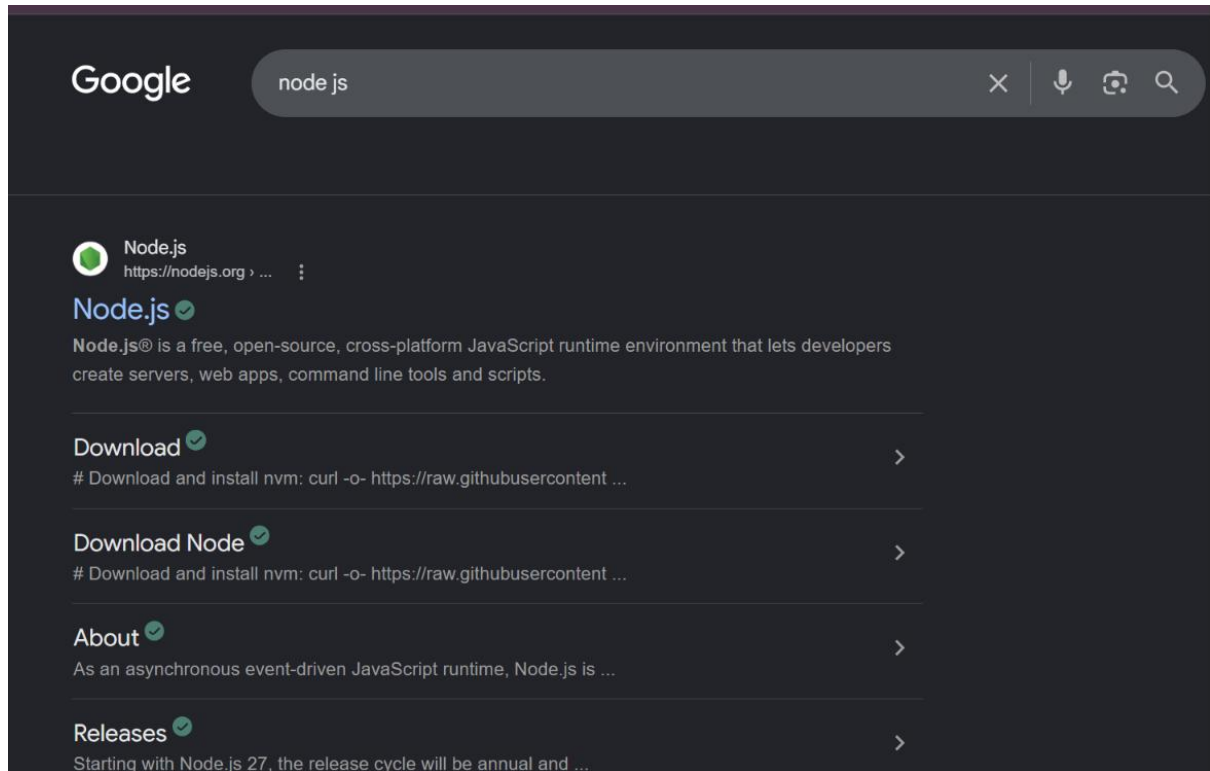


ReactJS Setup

Step 1:

Install Node



node.js Learn About Download Blog Docs Beta Docs Contribute Courses

Search Start typing...

Download Node.js®

Get Node.js® v24.18.1 LTS for Windows using Docker with npm

Info: Want new features sooner? Get the latest Node.js version instead and try the latest improvements!

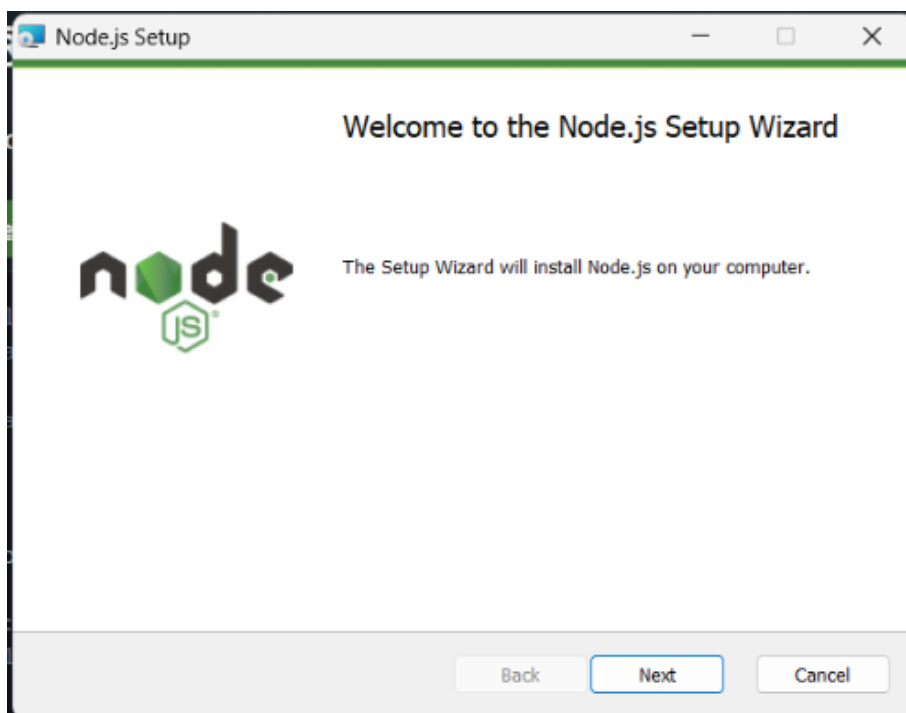
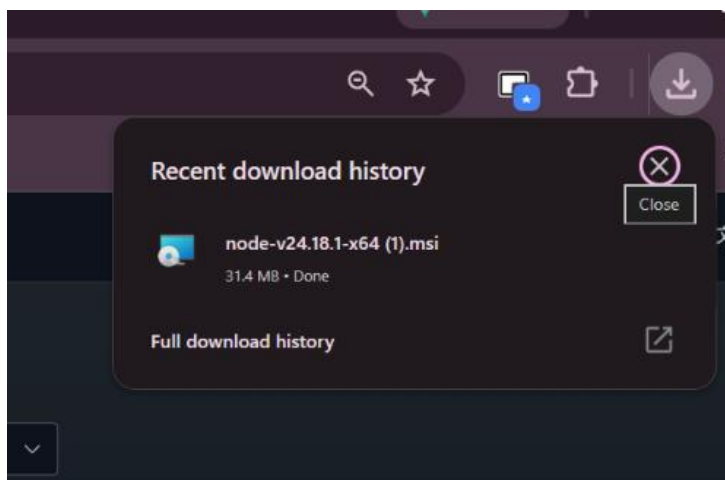
```
1 # Docker has specific installation instructions for each operating system.
2 # Please refer to the official documentation at https://docker.com/get-started/
3
4 # Pull the Node.js Docker image:
5 docker pull node:24-slim
6
7 # Create a Node.js container and start a Shell session:
8 docker run -it --rm --entrypoint sh node:24-slim
9
10 # Verify the Node.js version:
11 node -v # Should print "v24.18.1".
12
13 # Verify npm version:
14 npm -v # Should print "11.16.0".
```

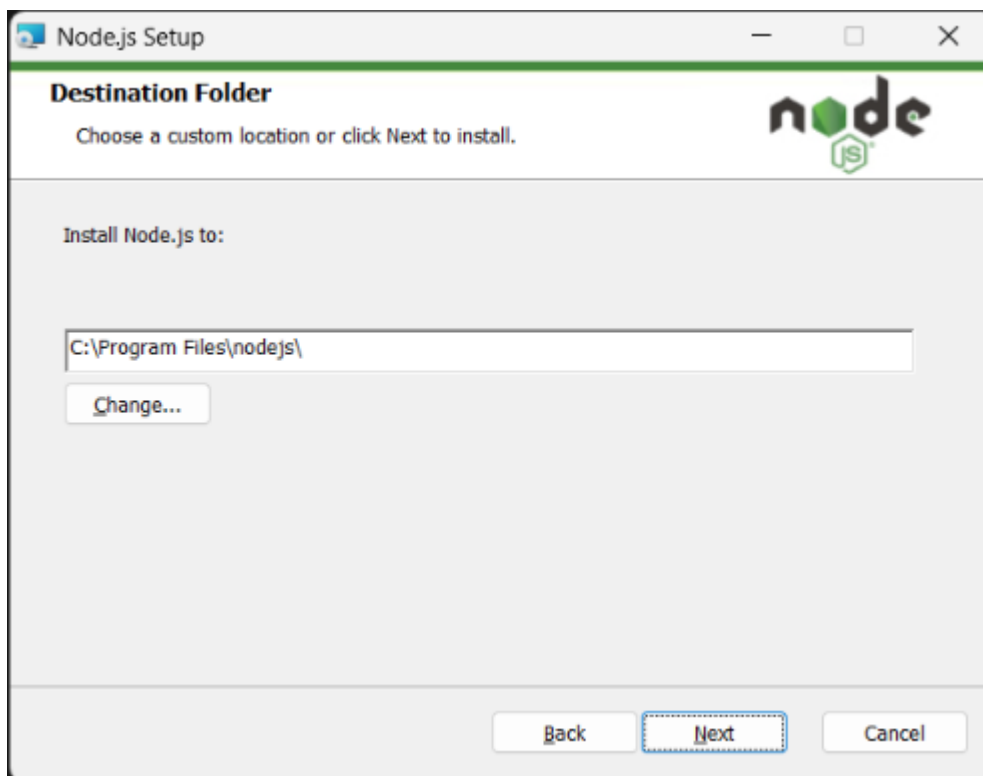
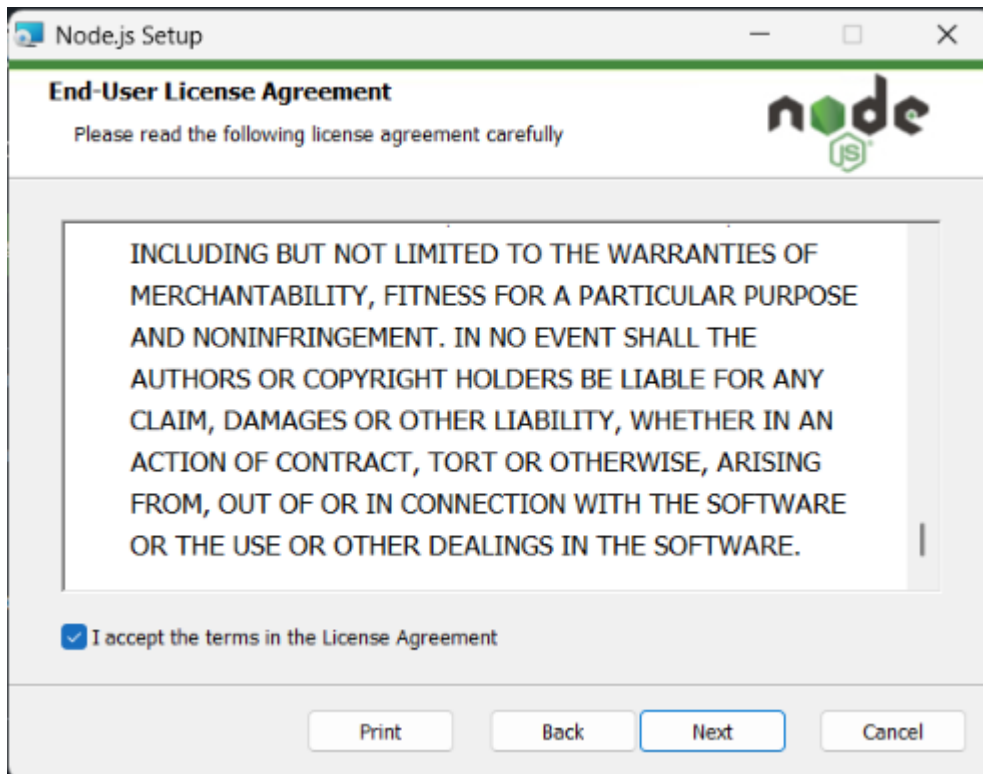
PowerShell [Copy to clipboard](#)

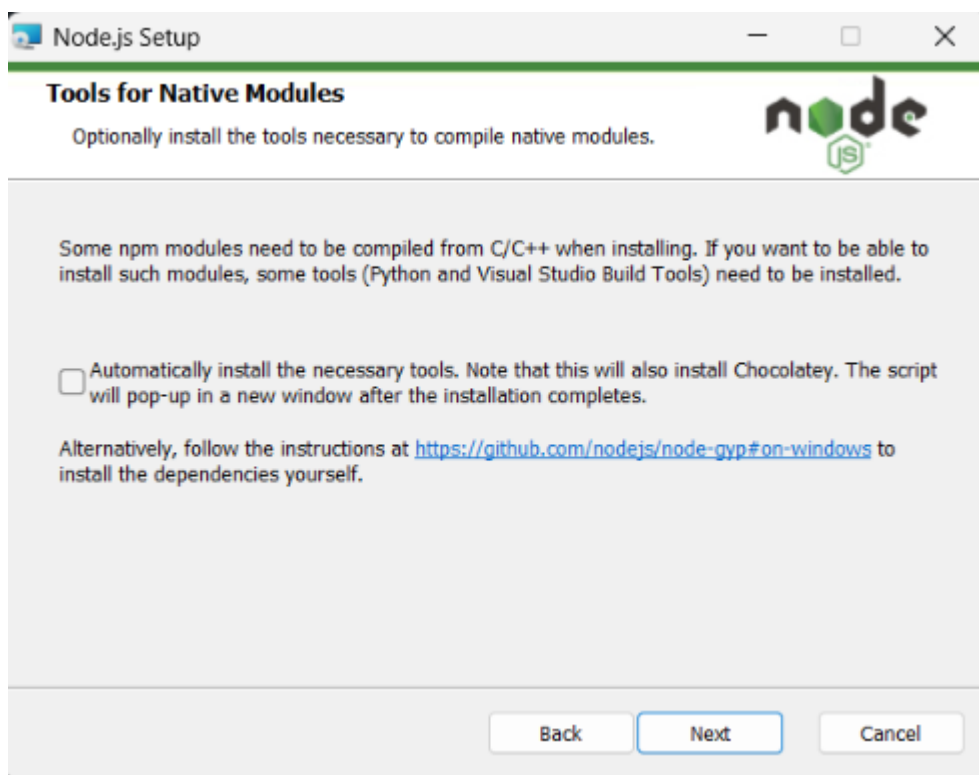
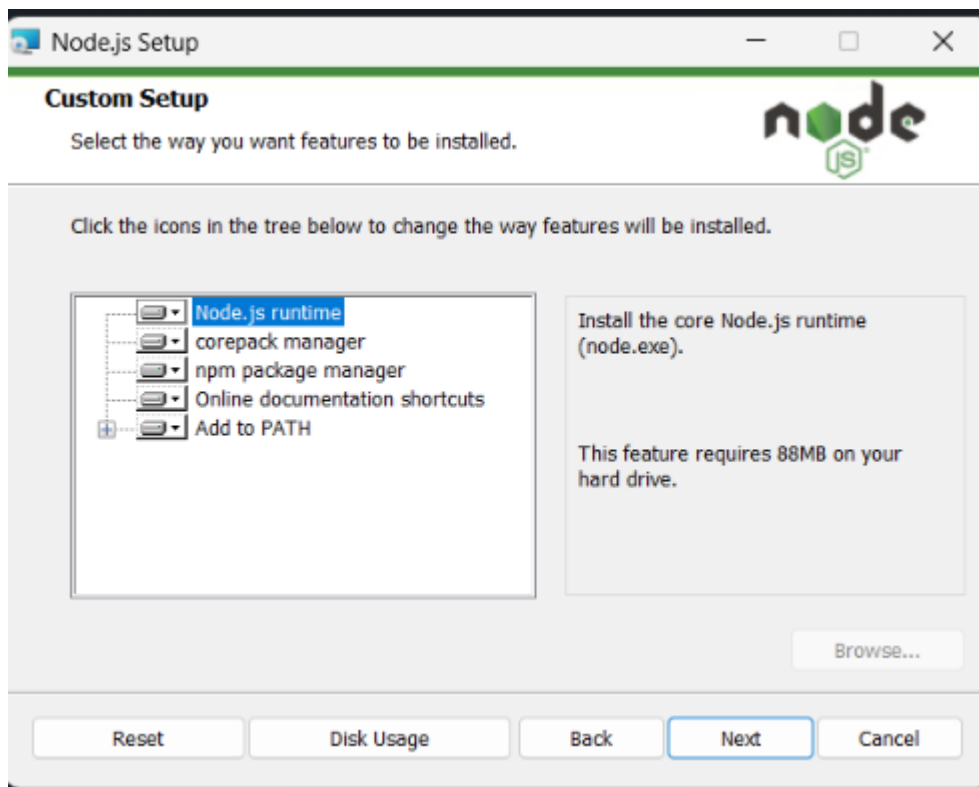
Docker is a containerization platform. If you encounter any issues please visit [Docker's website](#).

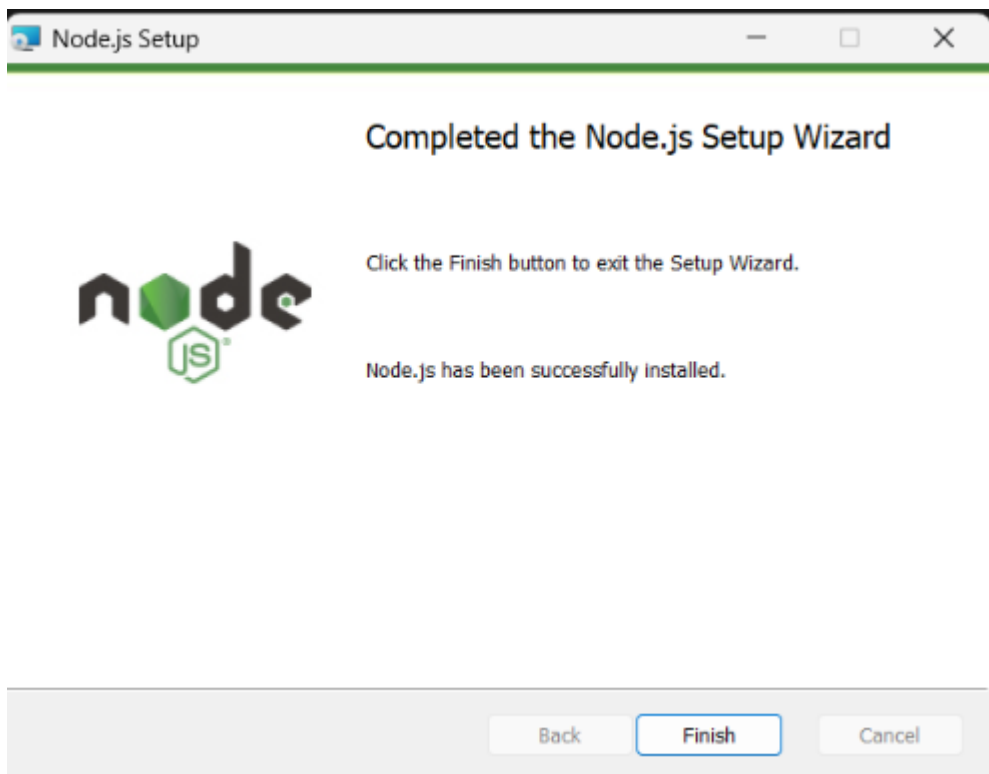
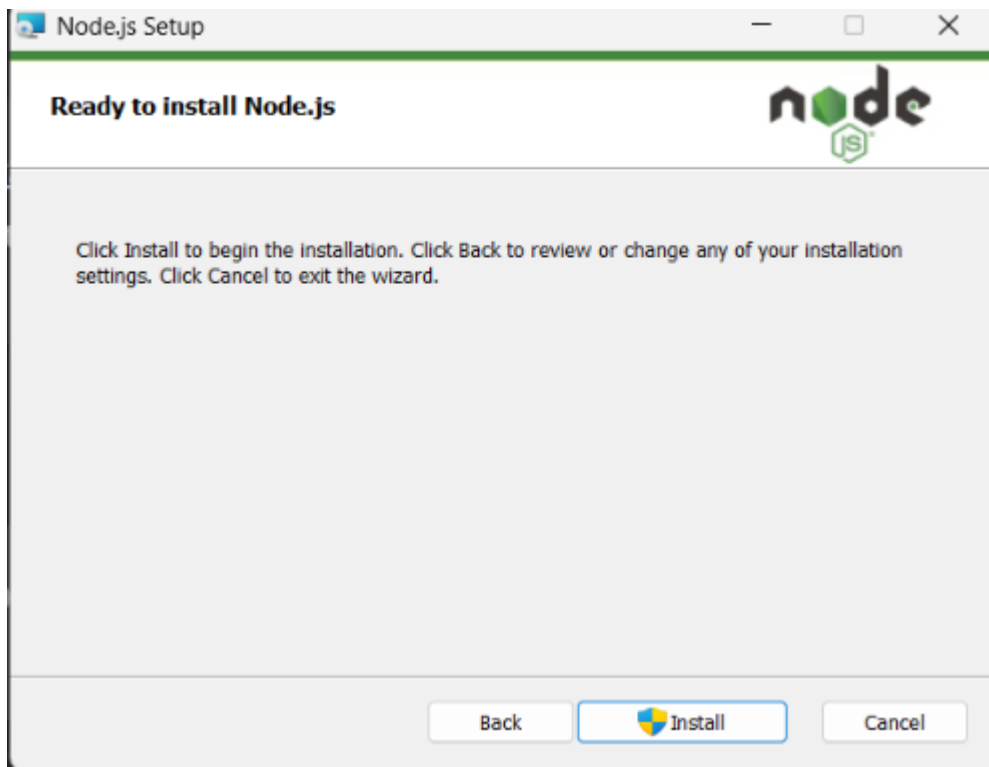
Or get a prebuilt Node.js® for Windows running a x64 architecture.

[Windows Installer \(.msi\)](#) [Standalone Binary \(.zip\)](#)









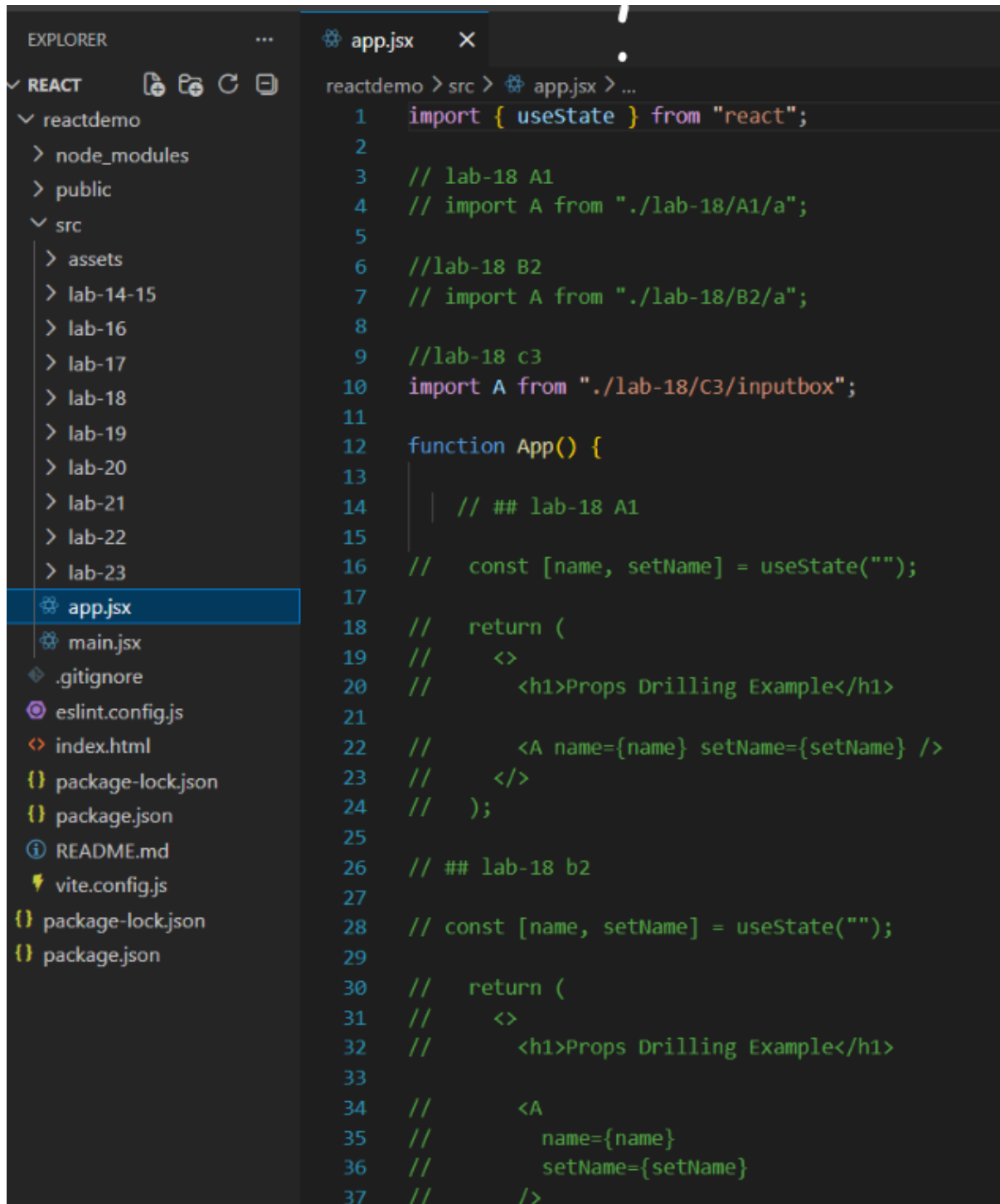
Check if its installed in cmd by : node -v command

```
C:\Users\sgn07>node -v  
v22.17.0
```

Step2 Installation of ReactJS

(Disclaimer: Throughout the semester we will use this projectFolder. Every labs from now on will be performed and implemented in this folder.

This will be your folder structure in the ProjectFolder that we create for lab work.



The image shows a screenshot of a code editor (VS Code) with a dark theme. On the left, the 'EXPLORER' sidebar displays a file tree for a project named 'reactdemo'. The tree structure is as follows:

- reactdemo
 - node_modules
 - public
 - src
 - assets
 - lab-14-15
 - lab-16
 - lab-17
 - lab-18
 - lab-19
 - lab-20
 - lab-21
 - lab-22
 - lab-23
 - app.jsx** (selected)
 - main.jsx
 - .gitignore
 - eslint.config.js
 - index.html
 - package-lock.json
 - package.json
 - README.md
 - vite.config.js
 - package-lock.json
 - package.json

The main editor area shows the content of 'app.jsx' with the following code:

```
1  import { useState } from "react";
2
3  // lab-18 A1
4  // import A from "../lab-18/A1/a";
5
6  //lab-18 B2
7  // import A from "../lab-18/B2/a";
8
9  //lab-18 c3
10 import A from "../lab-18/C3/inputbox";
11
12 function App() {
13   // ## lab-18 A1
14   // const [name, setName] = useState("");
15
16   // return (
17   //   <>
18   //     <h1>Props Drilling Example</h1>
19   //     <A name={name} setName={setName} />
20   //   </>
21   // );
22
23   // ## lab-18 b2
24   // const [name, setName] = useState("");
25
26   // return (
27   //   <>
28   //     <h1>Props Drilling Example</h1>
29   //     <A
30   //       name={name}
31   //       setName={setName}
32   //     />
33   //   </>
34   // );
```

).

Installation by Vite(Recommended)

```
npm create vite@latest
```

```
F:\TA (FEWT)\Lab Solution\react intro>npm create vite@latest
Need to install the following packages:
```

```
create-vite@9.1.2
```

```
Ok to proceed? (y) y
```

```
> npx
```

```
> create-vite
```

```
|
o Project name:
| rohaan-react
|
* Select a framework:
|  Vanilla
|  Vue
| > React
|  Preact
|  Lit
|  Svelte
|  Solid
|  Ember
|  Qwik
|  Angular
|  Marko
|  Others
| ↑/↓ to navigate • Enter: confirm
|
```

```
|
* Which linter to use?
|  Oxlint
| > ESLint
| ↑/↓ to navigate • Enter: confirm
|
```

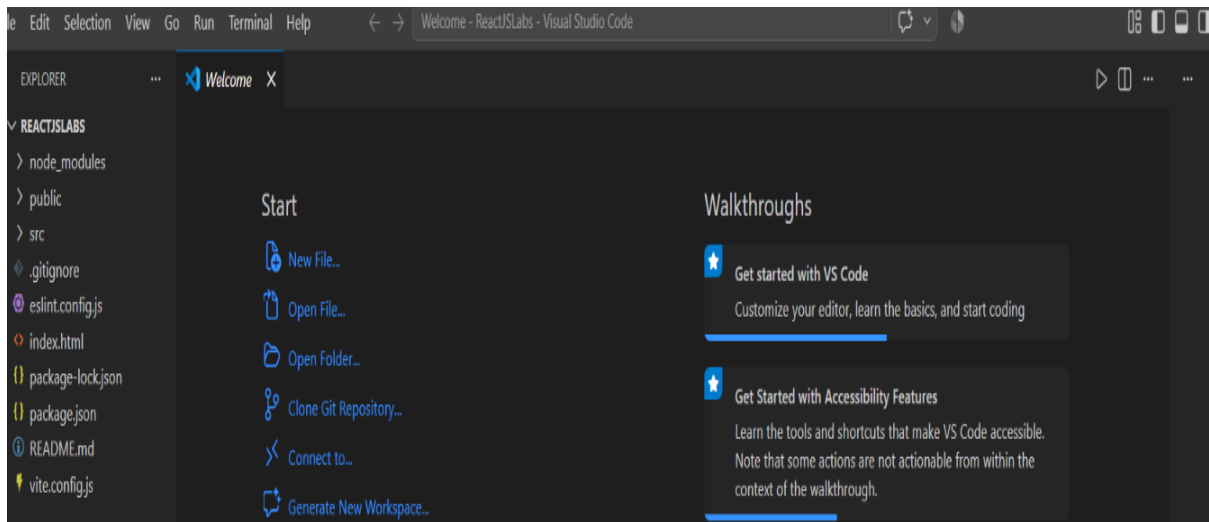
```
* Select a variant:
|   TypeScript
|   TypeScript + React Compiler
| > JavaScript
|   JavaScript + React Compiler
|   RSC
|   React Router v7 ↗ https://reactrouter.com
|   TanStack Router ↗ https://tanstack.com/router
|   RedwoodSDK ↗ https://rwsdk.com
|   Vike ↗ https://vike.dev
| ↑/↓ to navigate • Enter: confirm
```

```
* Install with npm and start now?
| > Yes / No
```

Command to stop the localhost: Ctrl + C (in cmd)

To start localhost:

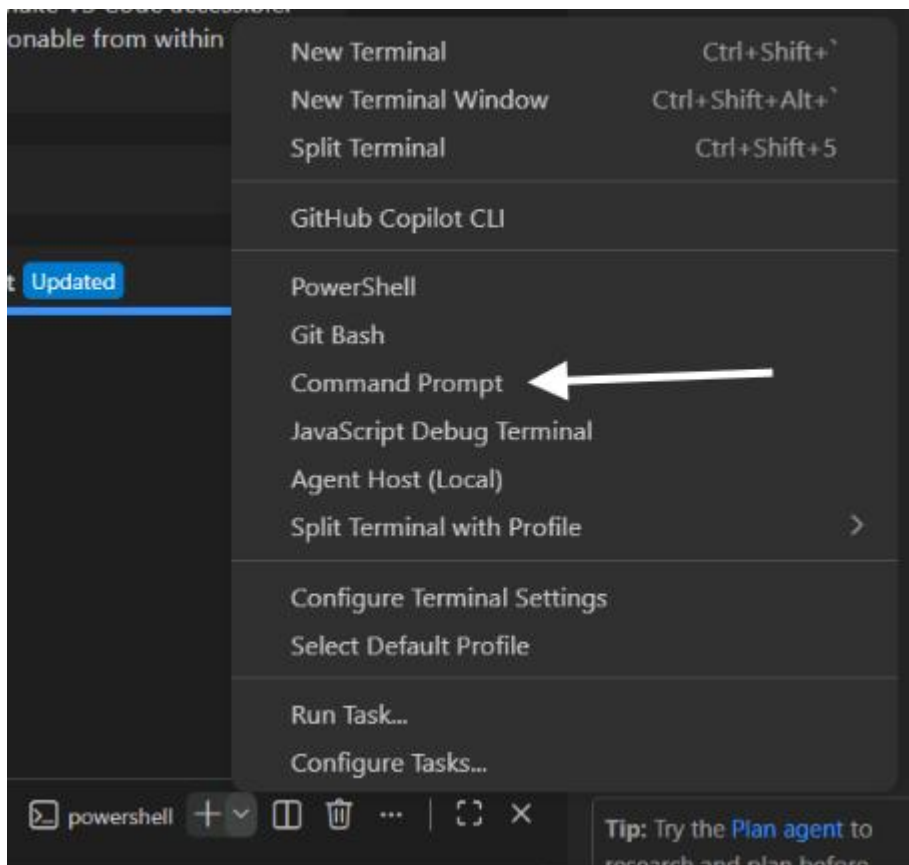
Go to the project folder:



Open the terminal(shortcut: CTRL + J)



Switch powershell to command prompt:



Run the following command: `ReactJS Labs>npm run dev`

```

> reactjslabs@0.0.0 dev
> vite

VITE v8.2.0 ready in 256 ms

→ Local:   http://localhost:5173/
→ Network: use --host to expose
→ press h + enter to show help

```

←Output

React Installation without Vite(W)

```

Happy hacking!

F:\TA (FEWT)\Lab Solution\react intro>cd rohaan-react-app

F:\TA (FEWT)\Lab Solution\react intro\rohaan-react-app>npm start|

F:\TA (FEWT)\Lab Solution\react intro>npx create-react-app rohaan-react-app|

Happy hacking!

F:\TA (FEWT)\Lab Solution\react intro>cd rohaan-react-app

F:\TA (FEWT)\Lab Solution\react intro\rohaan-react-app>npm start

> rohaan-react-app@0.1.0 start
> react-scripts start

(node:35072) [DEP0176] DeprecationWarning: fs.F_OK is deprecated, use fs.constants.F_OK i
(Use 'node --trace-deprecation ...' to show where the warning was created)
(node:35072) [DEP_WEBPACK_DEV_SERVER_ON_AFTER_SETUP_MIDDLEWARE] DeprecationWarning: 'onAf
ption is deprecated. Please use the 'setupMiddlewares' option.
(node:35072) [DEP_WEBPACK_DEV_SERVER_ON_BEFORE_SETUP_MIDDLEWARE] DeprecationWarning: 'onB
option is deprecated. Please use the 'setupMiddlewares' option.
Starting the development server...
Compiled successfully!

You can now view rohaan-react-app in the browser.

Local:      http://localhost:3000
On Your Network:  http://192.168.1.4:3000

Note that the development build is not optimized.
To create a production build, use npm run build.

webpack compiled successfully

```

What is ReactJS?

React JS is a Javascript Library used majorly for frontend development.

How does it build?

It builds using **Components**. Components are UI elements of the project.

Example: Your Project: Movie Review Database Frontend.

Components: Header, MovieSlider, MovieItems, Trailers, LatestNews and Footer

How does it work?

Basically it **doesn't** change the Document Object Model (DOM) in the browser directly.

Instead It creates a Virtual DOM and then updates the original DOM only the amnt it is needed.

React only changes what needs to be changed.

What is .jsx?

JSX stands for Javascript XML. It helps us to write HTML in a way that ReactJS accepts it.

```
const demoElement = <h1>This is only possible if we use JSX </h1>;
```

otherwise we will have to:

```
const demoElement = React.createElement('h1', {}, 'This does not use  
jsx so we have to explicitly write this to convert in  
ReactUnderstandable format.');
```